Program: B Tech and MBA Tech Data Science				**Semester:** V	
Course: Design Experience II				**Code:**	
Teaching Scheme				**Evaluation Scheme**	
Lecture (Hours per week)	**Practical (Hours per week)**	**Tutorial (Hours per week)**	**Credit**	**Internal Continuous Assessment (ICA) (Marks - 50)**	**Term End Examinations (TEE) (Marks - 100)**
0	4	0	2	Marks Scaled to 50	Marks Scaled to 50

Pre-requisite: -

Course Objective

This course develops students' skills in human-centered design, interdisciplinary teamwork, and problem-solving through structured project work. Students will engage in stakeholder analysis, prototyping, testing, and iterative refinement while integrating engineering principles, creativity, and sustainability. Emphasizing collaboration and ethical responsibility, the course prepares students for industry and community-driven challenges by enhancing their project management, communication, and critical thinking abilities.

Course Outcomes

After completion of the course, students will be able to --

1. Critically evaluate complex, real-world challenges using systematic problem-solving approaches.
2. Lead and collaborate in interdisciplinary teams to develop scalable and impactful solutions.
3. Apply advanced design methodologies and emerging technologies to prototype and refine innovative solutions in research.
4. Integrate sustainability, ethics, and feasibility analysis into the design and implementation process.
5. Demonstrate professional communication through technical reports, presentations, and stakeholder engagement.

Detailed Syllabus

Unit	**Description**	**Duration**
1.	The course covers Problem Analysis, where students assess the problem context, constraints, stakeholders, and solution stage while conducting literature reviews and benchmarking existing solutions. In Design Thinking & Concept Development, they apply the phases of Design Thinking—Empathize, Define, Ideate, Prototype, and Test—to develop and refine innovative solutions. Prototyping & Testing involves transforming ideas into practical prototypes, conducting experiments, usability testing, and iterative refinements. Through Project Management & Documentation, students assign roles, track progress, and maintain structured records. Communication & Review focuses on preparing presentations, conducting design reviews, and incorporating feedback. Finally, Reflection & Improvement allows students to analyze challenges, enhance their skills, and document key learnings.	**60**
	Total	**60**



Signature
(Head of the Department)

Laboratory Work: The Laboratory work comprises various activities performed by the project teams for formulation, solution implementation and modification. Each team needs to submit the following documents:

- Problem Statement Document
- Project Development Report
- Mid Semester Project Report
- Research Paper / Conference Paper
- Final Report

_____

Signature
(Head of the Department)



Signature
(Head of the Department)